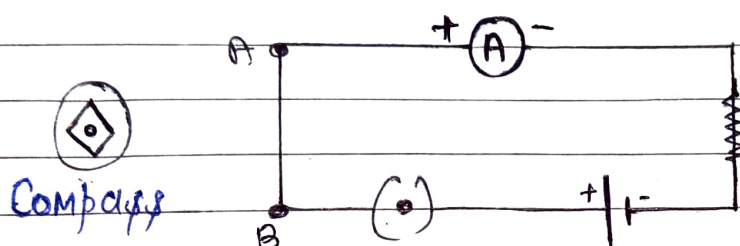


Chapter:→

Magnetic effects of Electric Current.

* Magnetic for needle deflection experiment :→

→ It is performed by Hans Christian Oersted (1820)



➤ Observations :→

→ When no current is flowing in the conductor AB, no deflection is shown in the magnetic needle.

→ If the direction of current is reverse, then compass needle shows deflection in the opposite direction.

→ If the compass is brought near the conductor then the compass needle shows more deflection.

• Conclusion :→

→ Current flowing in a conductor produces magnetic field around it and behave as a magnet.

→ current carrying conductor only attract or repel another magnet when it is in its magnetic field.

Magnet: →

MAGNET is an object which attract a piece of Iron (Fe), Nickel (Ni), and Cobalt (Co).

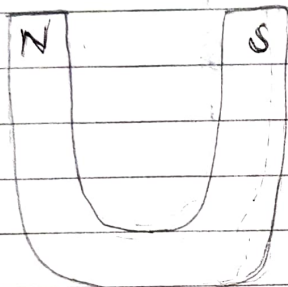


figure: → horse shoe magnet



figure: → bar magnet

* Properties of MAGNET: →

- (a) Every magnet has two poles namely North pole and south pole.
- (b) Unlike poles attract each other.
- (c) Like poles repel each other.
- (d) If a bar magnet is freely suspended then it aligns itself in the geographical north and south pole of the Earth.

➤ Magnetic Field (B) :- ➔

A space surrounding a magnet in which it shows attraction or repulsion is called

Magnetic field

→ It is denoted by the symbol $[B]$.

→ SI unit of magnetic field is Tesla (T).

• Magnetic field lines :- ➔

The imaginary lines drawn in a magnetic field are called Magnetic field lines.

Magnetic field lines due to a bar magnet :- ➔

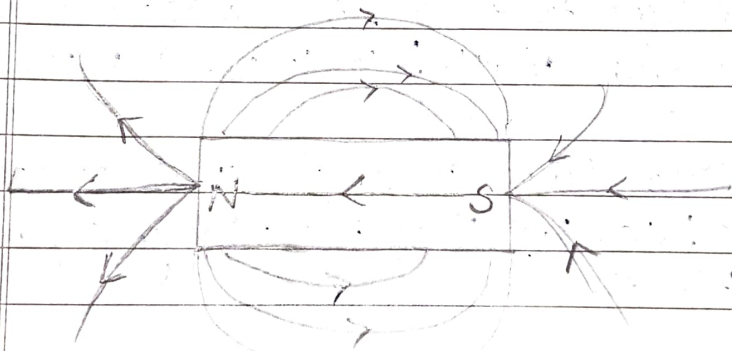


figure :- ➔ Magnetic field lines due to a bar magnet

→ direction of magnetic field lines outside the bar magnet is from north pole to south pole.

→ Direction of magnetic field lines inside the bar magnet is from south pole to north pole.

Note → Hans Christian Oersted was the first to state that electric current has magnetic field.

* Properties of Magnetic field lines : →

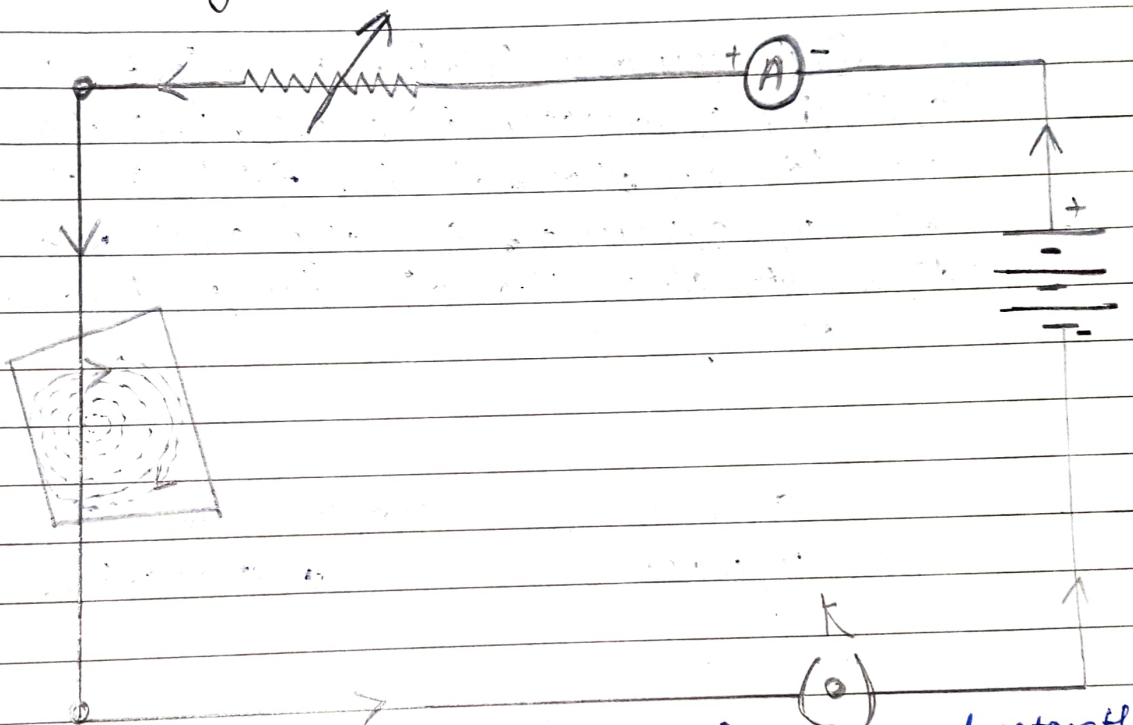
- ① Magnetic field lines originate from north pole and terminate at south pole outside the bar magnet.
- ② Magnetic field lines are closed curves.
- ③ Magnetic field lines are closer in stronger magnetic field.
- ④ Direction of magnetic field lines inside the bar magnet is from south pole to North pole.
- ⑤ Magnetic field lines come closer to one another near the poles but they are widely separated at other points.
- ⑥ **Imp.** Magnetic field lines never intersect each other.

Reason → If magnetic field lines intersect each other then at the point of intersection, they will show two directions which is not possible.

For text questⁿ (1) pg → 224.

questⁿ (1) → Compass needle is itself a tiny magnet when it is brought near the bar magnet then in the magnetic field of bar magnet, comp-
-ass needle shows deflection.

➤ Magnetic field due to a current carrying straight conductor →



→ In a straight current carrying conductor the pattern of magnetic field lines are concentric circles.

→ Nature of magnetic field is circular.

→ The centre of co-centric circles is wire.

→ Its direction can be find with the Maxwell right hand thumb rule.

Important points $\Rightarrow$

$\rightarrow$ Magnetic field $\propto$ strength of current

$\rightarrow$ Magnetic field $\propto \frac{1}{\text{distance from wire}}$

* Right hand Thumb rule [Maxwell Corkscrew Rule] $\Rightarrow$

Imagine you are holding a straight current carrying conductor in your right hand in such a way that, thumb points in the direction of current, then the curl of the fingers tells us about the direction of Magnetic field.



current
upwards

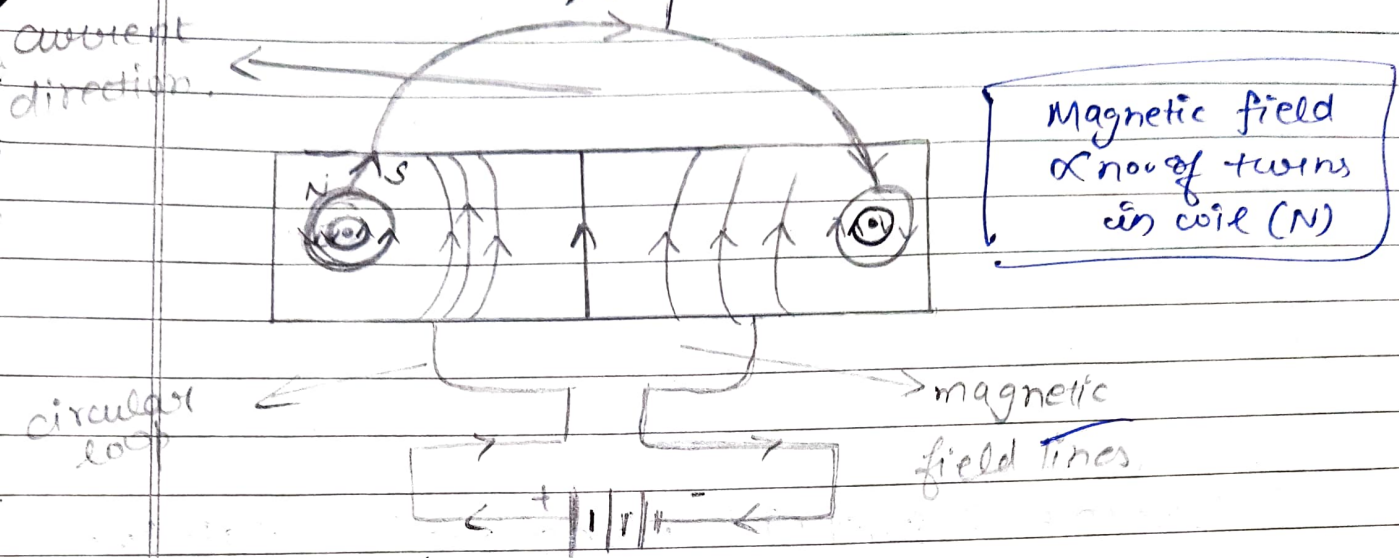
magnetic field direction $\Rightarrow$
Anticlockwise.



current
downwards

magnetic field direction $\Rightarrow$
clockwise.

➤ Magnetic field due to a current carrying circular loop



➔ Pattern of magnetic field lines are co-centric circles near the wire.

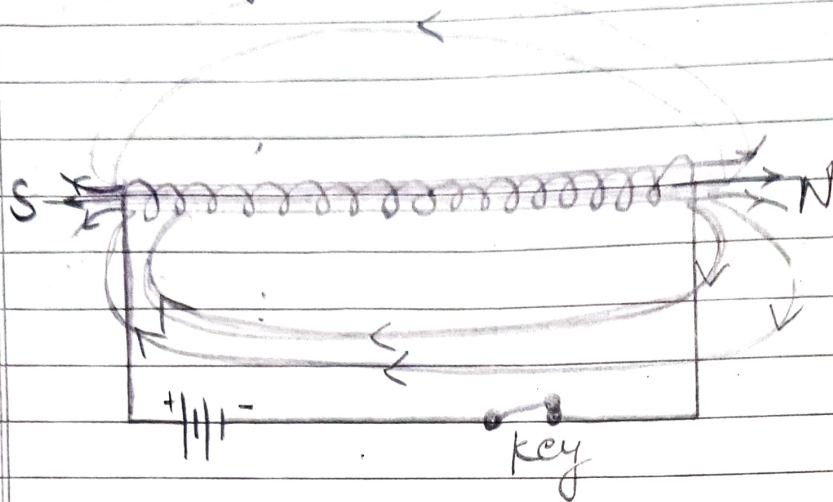
➔ As we move away from the wire, the lines become bigger and bigger.

➔ Magnetic field lines at the centre of loop is straight line.

➔ Magnetic field $\propto$ strength of current

➔ Magnetic field $\propto \frac{1}{\text{distance from wire}}$

➤ Magnetic field due to a solenoid: →



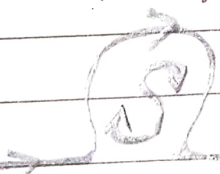
SOLENOID → solenoid a long coil containing a large no. of turns of insulated copper wire.

- Magnetic field of a solenoid is similar to that of bar magnet.
- Magnetic field outside the solenoid is from north to south pole.
- Magnetic field pattern inside the solenoid is from south to north pole.
- Impo Magnetic field inside the solenoid is uniform magnetic field.

Some Important points :-

- Magnetic field $\propto$ strength of current
- Magnetic field $\propto$ no. of turns (n).
- Magnetic field also depend on nature of core material inside the solenoid.
- Solenoid can be used to magnetise a magnetic material like soft iron.
Reason $\rightarrow$
 - soft iron produce stronger magnetism.
 - Also it loses its magnetism when current is switched off.
 - Soft iron is considered as the best core material for solenoid.

* Clock face rule $\rightarrow$



current
clockwise



current
anticlockwise.

• Differentiate b/w Electromagnet & Permanent

Electromagnet:

Permanent Magnet

1. It is a temporary magnet so it can be easily demagnetised.

cannot be easily demagnetised.

2. Strength can be changed

strength is fixed.

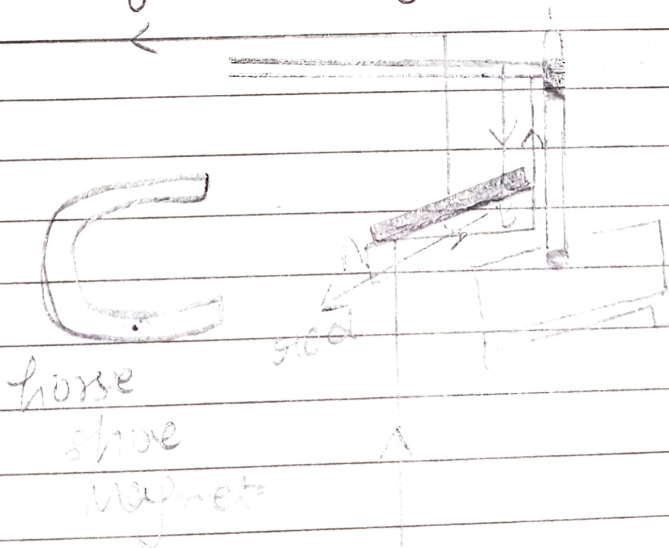
3. Polarity can be reversed.

Polarity cannot be reversed.

4. Generally strong magnet

Generally weak magnet

* Force on current carrying conductor placed in a magnetic field. (Kicking wire experiment) →



- It is performed by Andre Marie Ampere.
- According to him, when a current carrying conductor is placed in a vicinity of magnet exerts magnetic force on rod which causes rod to move or motion of a rod.

Important points: →

1. $F = BIL \sin \theta$

F = magnetic force

B = magnetic field

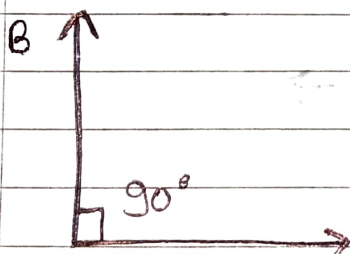
I = current

L = length of rod (conductor)

θ = angle b/w magnetic field and current.

Impo

→ Displacement in the conductor (rod) is maximum when current is at right angle to magnetic field.



If $\sin \theta = 0^\circ$ [minimum displacement]

If $\sin \theta = 90^\circ$ [maximum displacement]

minimum force at $\theta = 0^\circ$

← A → B Then Anti Parallel

- If the direction of current is reversed then rod shows displacement in opposite direction.

◦ Hemming's left hand rule : →

↳ flam used to find magnetic force direction.

→ It is used to find the direction of magnetic force.

→ Hold your left hand in such a way that your thumb, forefinger, centre finger

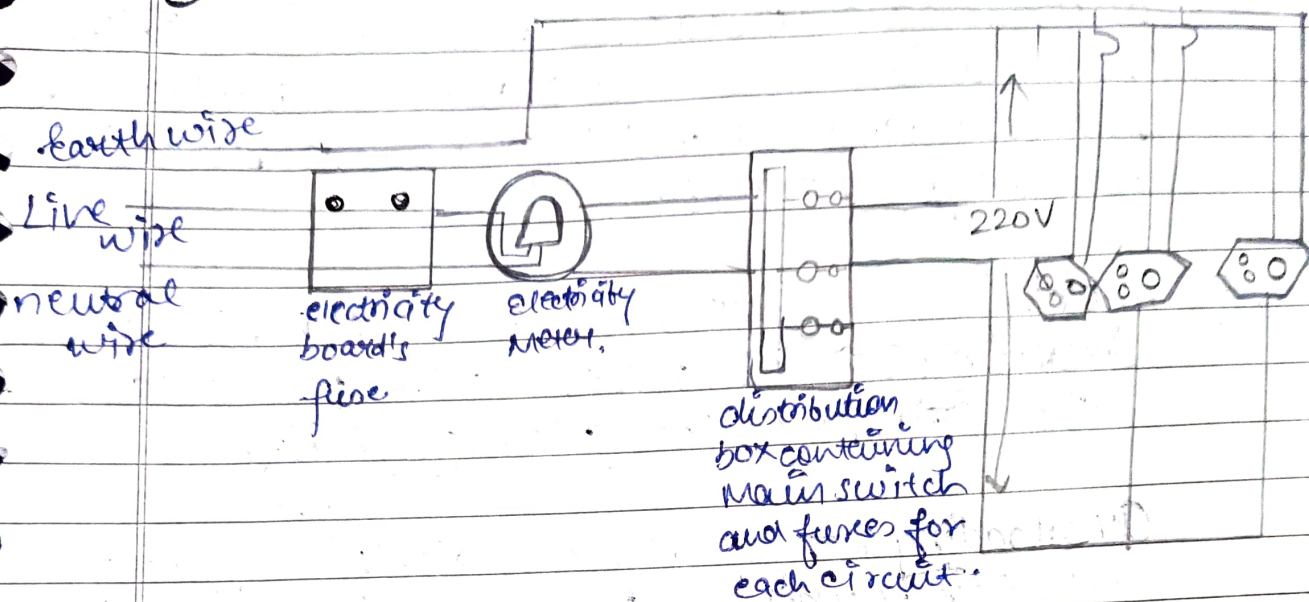
→ Hold your thumb, forefinger and centre finger of your left hand in such a way that forefinger points in the direction of current then thumb of your left hand points in the direction of magnetic force.

TRICK : → FBI
→ FMC

father mother child
(force) (B) (current)

➤ Domestic Electric Circuit : →

* DOMESTIC ELECTRIC CIRCUIT :-



① Pole → Main supply → fuse → electric meter. →
distribution box → To separate circuits.

There are three kinds of wire :-

- ① Live wire (+ve) with red insulation cover.
- ② neutral wire (-ve) with black insulation cover.
- ③ Earth wire (neutral) with green insulation cover.

→ The potential difference b/w the live wire and neutral wire in India is 220 volt

- **Earth** : $\rightarrow$ It protects us from electric shock wire in case of leakage of current especially in metallic body appliances. It provides a low resistance path for current in case of leakage of current.
- **Short circuiting** : $\rightarrow$ When live wire comes in contact with neutral wire accidentally then it is called **short circuiting**.
- **Overloading** : $\rightarrow$ If too many electric appliances of high power rating are switched on at the same time this is known as **overloading**.

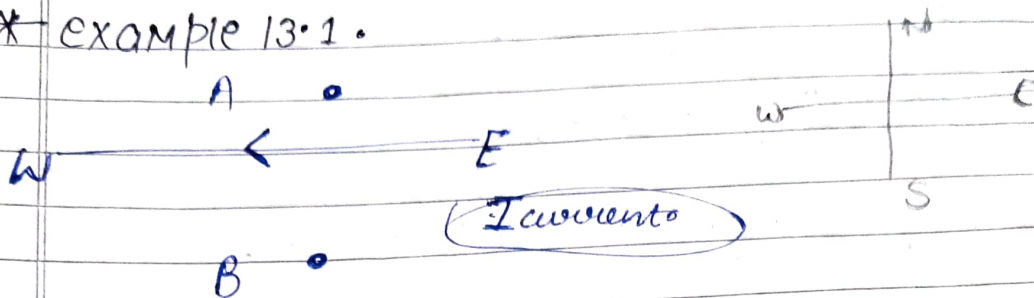
* **Causes of overloading** : $\rightarrow$

- ① **Accidental hike in voltage supply**
- ② **Use of more than one device in single socket.**

• **Safety devices** : $\rightarrow$

- ① **MCB [Miniature Circuit Breaker]**
- ② **Fuse**
- ③ **Earth wire.**

* example 13.1.



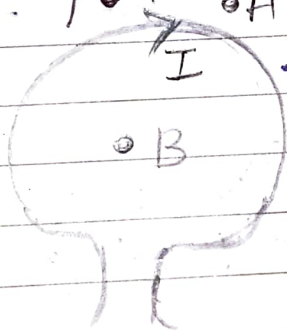
Lawrence

→ At point above the horizontal line direction of magnetic field is into the page or plane

→ At point below the horizontal line direction of magnetic field is out of the page or plane

Inte xt questⁿ : → pg no → 229

questⁿ (1) →



→ At the point inside the loop the direction of magnetic field is into the plane of table.

→ At the point outside the loop the direction of magnetic field is out of plane of table.

questⁿ (2) →

questⁿ (3) → is the same at all points.

pg $\rightarrow$ 232

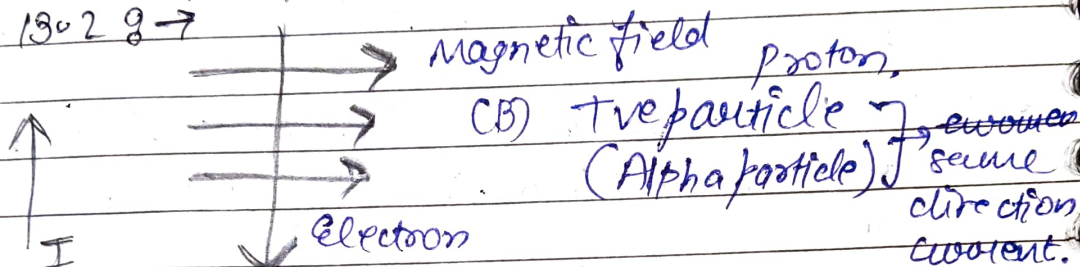
displacement is

questⁿ (2) $\rightarrow$ cis° increase.

cis Magnetic field will be increase.

cis increase.

example 13.2 $\rightarrow$



Fleming's left hand rule.

Electron $\rightarrow$ opp. direction current.

(d) $\rightarrow$ Into the page.

pg $\rightarrow$ 231.

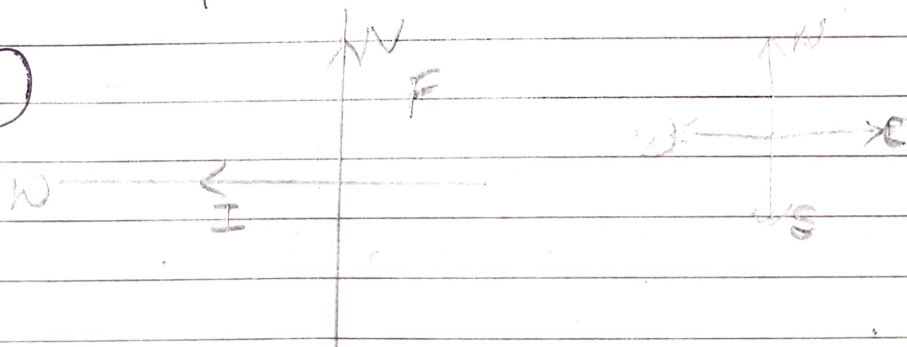
Interact questⁿ (1) $\rightarrow$ +ve particle.

$\rightarrow$ velocity, momentum (e) $\rightarrow$ both (c) and (d).

$$\text{momentum (P)} = m \times v$$

velocity $\rightarrow$ vector (M + D).

questⁿ (3)



(d) $\rightarrow$ upwards.

For text questⁿ → pg → 238.

questⁿ → 1 → earthing wire & electric fuse.

(2) → $P = 2 \text{ kW} = 2 \times 1000$
 $= 2000 \text{ W.}$

[short circuit will happen]

$V = 220 \text{ V}$

$I = 5 \text{ Ampere.}$

$P = V \times I$
 $2000 = 220 \times I$

$\frac{100 \times 2000}{220} = I$
11

Result → The fuse of 5 Ampere will blow up. If there is no fuse in the circuit the appliance will get damage.

$\frac{100}{11} = I$
 $I = 9.09 \text{ Ampere.}$

exercise questⁿ → pg → 240.

questⁿ (1) → The field consists of concentric circles centred on the wire. (d) ✓.

(2) → (c) increases heavily ✓.

(3) → (a) True. (b) False.

(5) → Force experienced by the current carrying conductor placed in a magnetic field is largest when magnetic field and current are perpendicular to each other.

ques → (6) downwards.

→ direction of the magnetic field is downwards.

ques → (7) (i) Maxwell right-hand thumb rule.

(ii) Fleming's left hand rule.

(iii) Fleming's right hand rule.

ques → (8) When the live wire comes in contact with neutral wire accidentally then it is called short circuiting.

ques → (9) The metallic body of electric appliance is earthed by the use of earth wire. If any leakage of current occurs then the current is transferred into ground.

- It prevents us from electric shock in case of leakage of current.

ques → (7) • By using solenoid.

• By using circular loop.

• By using long straight wire carrying current.

• By using magnet.